

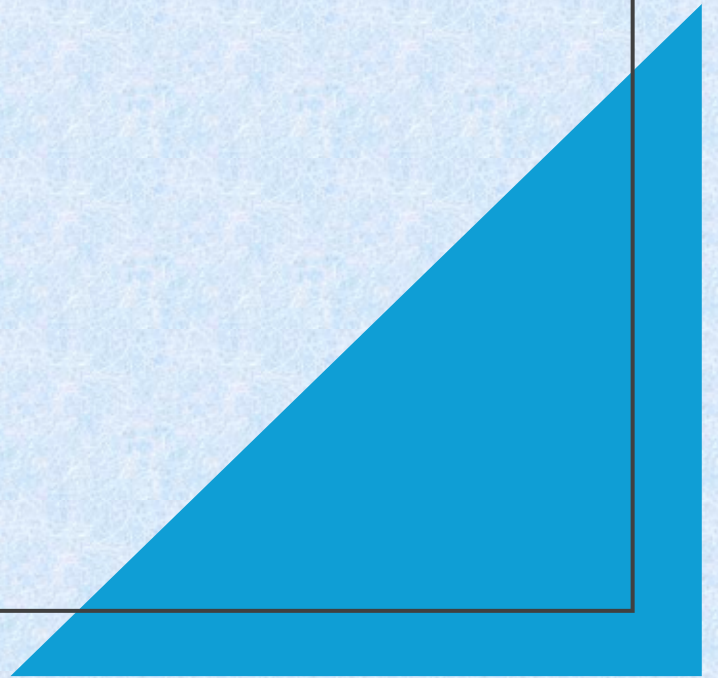
Technical and Business Writing

Course instructor: Ms. Mariam Aftab

mariam.aftab@nu.edu.pk



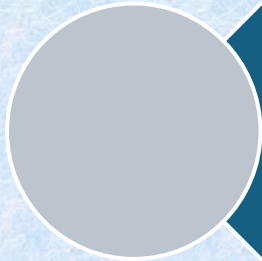
Project: Writing a Research Proposal



Choose your Nature of Project



Development



Research & Development



Research

Nature of Project

Research-Based (Bonus 2 weightage)

Focus:

Investigating a problem through existing research

Data source:

Research Papers

Development-Based

Focus:

Designing a product/system/app

Data source:

Existing products, tools, applications (survey)

Research and Development (Bonus 2 weightage)

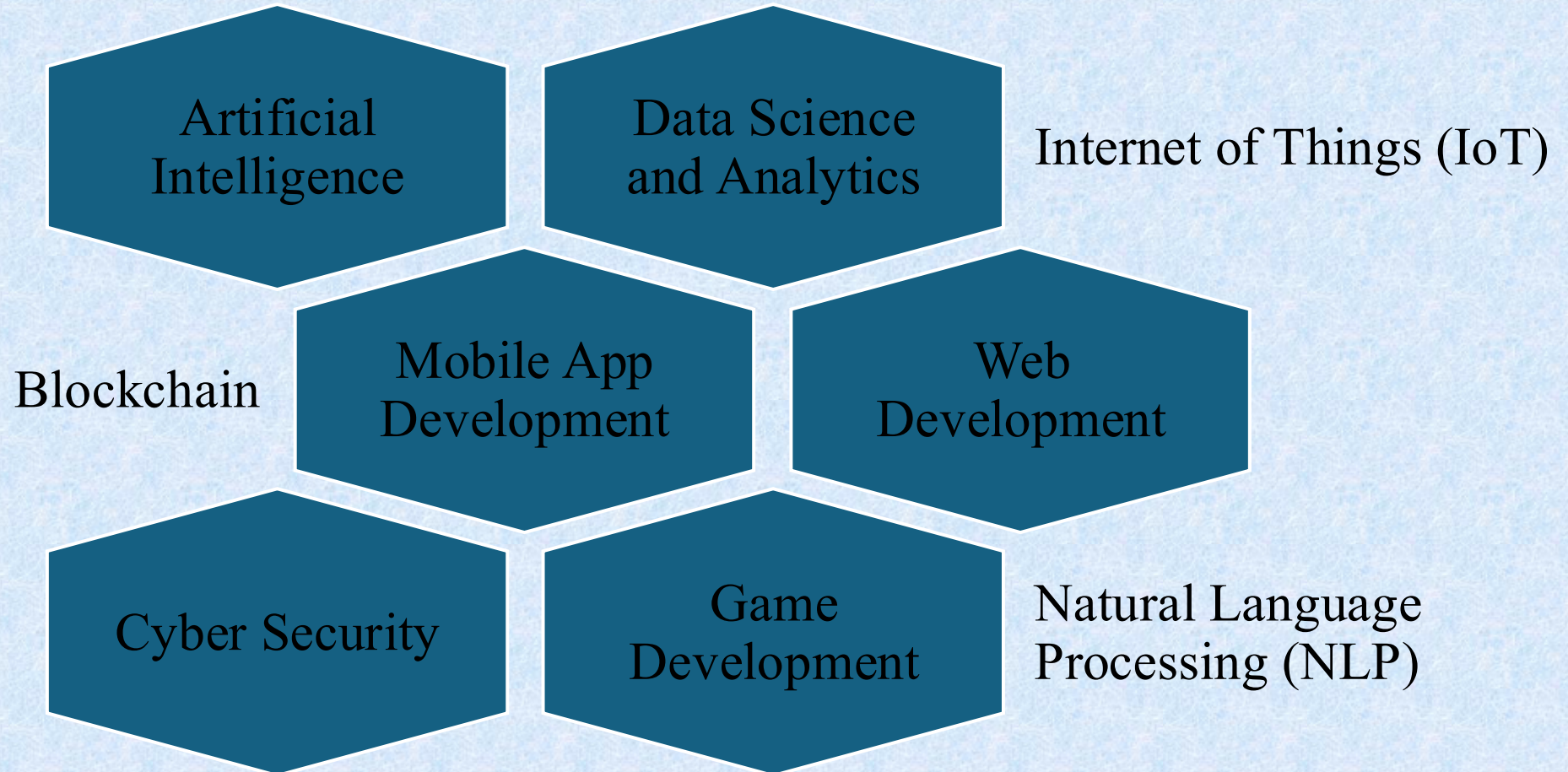
Focus:

Combines both research and product development

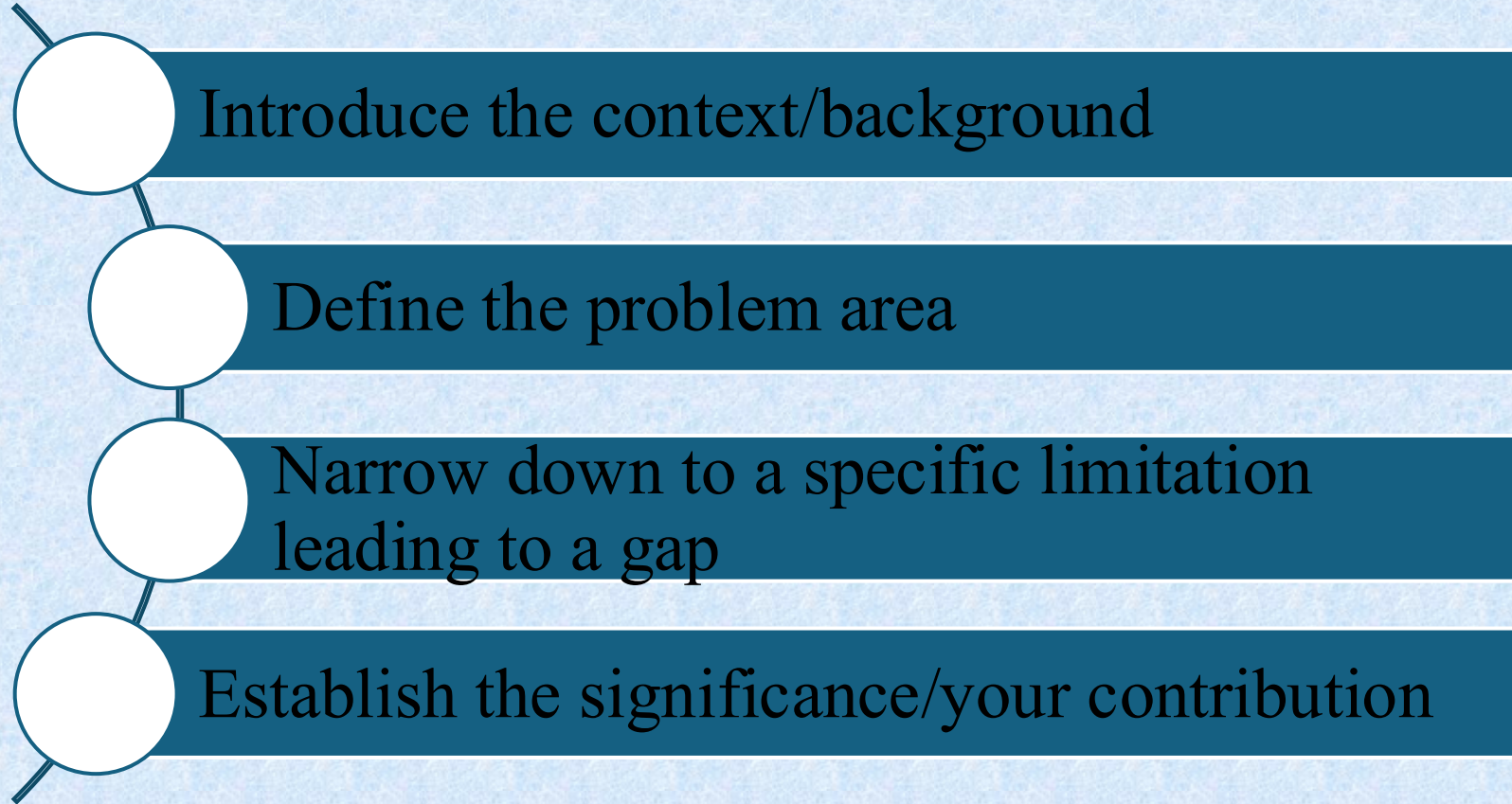
Data source:

Research papers + Existing tools/products

Choose your area of specialization



Structure of your proposal: Introduction



- ✓ Use citations when referring to statistics, or existing research/product.
- ✓ General statements may not require citation, but claims must be supported

Structure of your proposal: Related Work

For Research-Based Projects

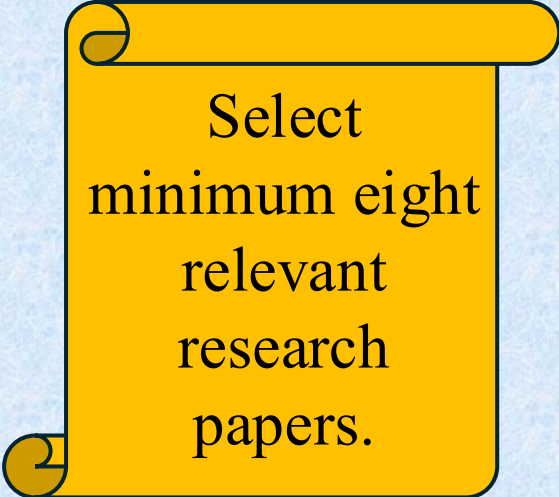
Source: Research Papers

Focus:

- What studies have already been conducted?
- What methods/techniques were used?
- What were the findings?
- What are the future work, limitations, and recommendation of these studies

You must:

- Synthesize studies, not summarize
- Compare studies (data/problem/methodology)
- Show patterns like similarities, agreements, disagreements, limitations

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Select
minimum eight
relevant
research
papers.

Structure of your proposal: Related Work

For Development-Based Projects

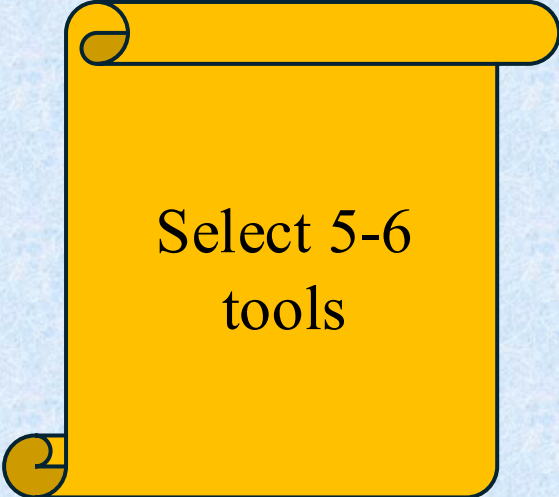
- Source: Existing products/apps/systems
- Focus:
- Features of similar tools & techniques
- Strengths and limitations
- Usability gaps

You must:

- Conduct a application/tool-based survey
- Compare multiple tools

Cite

- Official websites
- Product documentation

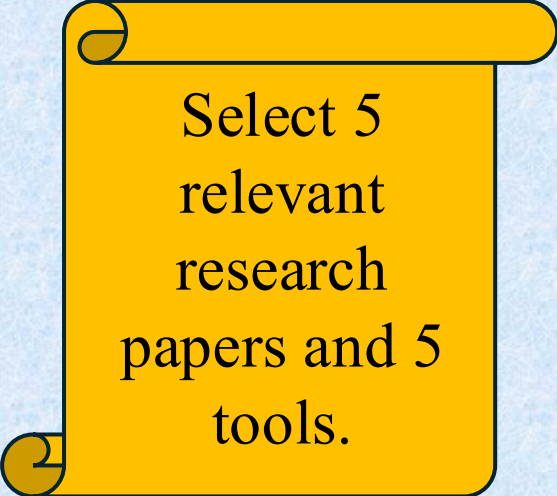


Select 5-6
tools

Structure of your proposal: Related Work

For Research & Development Projects

- Source: Research Papers + Product analysis
- Your review should answer:
- What does research say?
- What do current tools offer?
- What is still missing?

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Select 5
relevant
research
papers and 5
tools.

Publishers of quality research studies

- MDPI (<https://www.mdpi.com/>)
- Wiley (<https://www.wiley.com/en-us/>)
- Elsevier (<https://www.elsevier.com/>)
- Science direct (<https://www.sciencedirect.com/>)
- Springer (<https://link.springer.com/>)
- Taylor & Francis (<https://www.tandfonline.com/>)

Structure of Research Proposal: Research Gap

Research



- ☐ Limited sample size
- ☐ Limitations in methodology
- ☐ Contradictory findings

Development



- ☐ Missing features
- ☐ Poor usability
- ☐ Inefficiency
- ☐ Integrate new technology

Research + Development



- ☐ Mismatch between research findings and real-world applications
- ☐ Incomplete research work

Gap must be supported by your related work analysis

Structure of research proposal: Purpose and Objectives

Purpose

- Group of statements stating what your project aims to do

Objectives

- Specific, measurable steps align with your identified gap, problem statement and purpose of the study

Structure of research proposal: Methodology

- Briefly write about how you plan to carry out your project.

Structure of research proposal: Reference List

- Provide references of all the work you cite using IEEE format

Submission of Research Proposal

- Title page: Project name, nature of project, area of specialization, group members
- Following headings will be the part of your proposal
 - ✓ Introduction (250-300 words)
 - ✓ Related work (add sub-headings)
 - ✓ Research gap
 - ✓ Purpose
 - ✓ Objectives
 - ✓ Methodology
 - ✓ Reference list

Proposal Submission:
April 26th
Weightage: 5

Proposal Presentation:
Week 16
Weightage: 2